CS 501 E1 · PROJECT UPDATE 1 · NOVEMBER 10, 2026

# Habitable

A heat-and-repair record for Boston renters: our working vertical slice, from sensor to landlord notice to server.


TEAM

**Ankita Patra · Priyanshu Singh**

SCRUM MASTER THIS WEEK

**Ankita Patra**

THE VERTICAL SLICE

# One flow, end to end

1
### Sensor
Simulated now, Bluetooth next

2
### Room
Insert-only readings, one transaction each

3
### Rules
68 / 64 °F, Sept 15 to May 31

4
### Chart
Step limit plus Open-Meteo weather

5
### Issue
Opens itself after 2 hours cold

6
### Notice
Dated email letter to the landlord

7
### Sync
Readings and issues to our API

**Every step is real code. Nothing on screen is hard-coded.**

Compose UI → ViewModel → repository → Room, Open-Meteo and our Express API.

ABOUT FIVE MINUTES, ON THE EMULATOR

# Live demo

## Priyanshu drives

1. Set up the apartment (validated, saved)
2. Start monitoring: foreground service
3. Rotate: state survives
4. Replay a cold night: 97 readings
5. Go offline: weather from cache

## Ankita drives

1. Home turns red: 6.6° below the limit
2. Open the issue: days-open clock, Share
3. Send notice: email to the landlord
4. Request inspection; search the Issues list
5. Sync now: readings and issue on the server

ARCHITECTURE

# Room is the source of truth

**Compose UI**
7 destinations · type-safe Navigation Compose · stateless *Content composables

**ViewModels**
One StateFlow<UiState> per screen · combine + stateIn · survives rotation

**Repositories**
Readings · Weather · Issues · Apartment · Sync: the only code that touches data

**Room + DataStore**
7 tables · settings

**Open-Meteo**
Retrofit, cached

**Our API**
Express + JWT

## Domain, pure Kotlin

No Android imports, unit-tested at the legal boundaries

- HeatRules
- ViolationDetector
- DataQualityChecker
- WeatherCrossCheck
- NoticeTemplate

Sensor: foreground service → SensorSource (simulated or BLE) → ReadingsRepository

PERSISTENCE AND API EVIDENCE

# Local data, a third-party API and our own

## Room + DataStore

- 7 tables; readings are insert-only
- Flags, violations and auto-issue in one transaction
- Settings and sign-in token in DataStore
- Shown live in Database Inspector

## Open-Meteo

- Hourly outdoor temperature via Retrofit
- Cached in Room, so it works offline
- Loading, error and stale states with Retry
- Cross-check: is the sensor outdoors?

## Our Express API

- JWT sign-in, bcrypt passwords
- POST /readings, deduped by sensor and time
- PUT /issues, upserted by client id
- PostgreSQL, or in-memory for demos

**85**
automated tests, all passing

**66**
unit: rules, letters, sync, issue tools

**6**
on-device: Room pipeline, Compose UI

**13**
backend API

WORK STATUS

# Done, in progress, next

### DONE · UPDATE 1

- Setup with validation, saved
- Simulated sensor: live and replay
- Chart vs. step limit and weather
- Automatic heat issue, escalation
- Notice letter by email
- Readings and issue sync
- Issue search, share, days-open clock
- Redesigned UI

### IN PROGRESS

- Bluetooth thermometer on real hardware (blocked: device ordered)
- Backend deploy to Render + Neon
- Notices stored as server records

### NEXT · UPDATE 2, NOV 24

- Photos hashed at capture
- PDF evidence packet
- Reminders after certification
- Address geocoding
- Accessibility pass

BACKLOG, ROLES AND SCOPE

# Prioritised backlog

| ID | Story | Owner | Status |
|---|---|---|---|
| US-3 | Chart vs. legal limit | Priyanshu | Done |
| US-5 | Issues + automatic heat issue | Ankita | Done |
| US-6 | Notice + reminders | Ankita | Partial |
| US-1 | Account + apartment | Both | Done |
| US-10 | Open-Meteo weather | Priyanshu | Done |
| US-2 | BLE thermometer + photo | Priyanshu | Blocked |
| US-4 | Photos hashed at capture | Ankita | Update 2 |
| US-7 | PDF evidence packet | Ankita | Update 2 |
| US-9 | Backend deployed | Priyanshu | Partial |

SCRUM MASTER

## Ankita, week of Nov 10

Rotates weekly. Runs the board, Tuesday planning and Friday review.

SCOPE CHANGES

- Canvas chart instead of Vico (sign-off?)
- One cold night = one issue
- Notice sent from the tenant’s email app
- Geocoding moved to sprint 2

Nothing added or dropped.

RISKS AND BLOCKERS

# What could go wrong, and our plan

| Type | Risk | Mitigation |
|---|---|---|
| **Device: blocker** | Bluetooth thermometer not yet in hand | Decoder unit-tested; simulator uses the same pipeline |
| Device | Android stops Bluetooth sampling in the background | Foreground service; gaps are flagged, never hidden |
| Permissions | Notifications or Bluetooth denied | Asked only when needed; Bluetooth scan with neverForLocation |
| Privacy | Address, photos, landlord details are sensitive | Stay on the phone; only readings and an issue summary sync, after sign-in |
| API | Open-Meteo or our laptop backend is down | Weather cached with Retry; backend moves to Render + Neon |
| Usability | Legal text read as legal advice | “Documentation only, not legal advice” on screens and letters |
| Schedule | Camera and PDF are both large | Camera in sprint 2, PDF in sprint 3; stretch work frozen |

REPOSITORY EVIDENCE

# How we work

## Issues and project board

Every backlog story is a GitHub Issue with an owner, size and milestone, on a To do / In progress / Done board.

## Branches and pull requests

Since Nov 8 every change is a branch and a reviewed pull request linked to its issue: #19 docs and #20 CI fix (Priyanshu), #21 issue sync, search, share and clock (Ankita, 4 commits).

## Continuous integration

GitHub Actions on every push: backend tests, Android build and unit tests.

## Cadence and disclosure

Daily check-in; Tuesday planning, Friday review. AI use disclosed in docs/AI_USE.md.

> Lesson learned: Oct 20–24 we committed straight to main from one machine. Since Nov 8, main is protected and both of us work through pull requests.

PLAN TO UPDATE 2

# Feature-complete beta by Nov 24

SPRINT 2 · NOV 10–17

- CameraX photos, SHA-256 at capture
- Placement photo for the sensor
- Notices stored on the server
- Deploy backend, geocode address

SPRINT 3 · NOV 17–24

- On-device PDF evidence packet
- Reminders after certification
- TalkBack and font-size pass
- Device tests in CI

# Questions?

github.com/Priyanshusingh09876/Habitable